

APPASA Strategy Document

Stage 6 – Act: Strategic Business Recommendations & Action Planning



APPASA: Act Stage

Introduction

I will use this **APPASA strategy document** to record my decisions and reflections as I work through **Stage 6: Act** of this project. This document will serve as a guide, helping me consider my responses and reflections throughout this stage of the data analysis process.

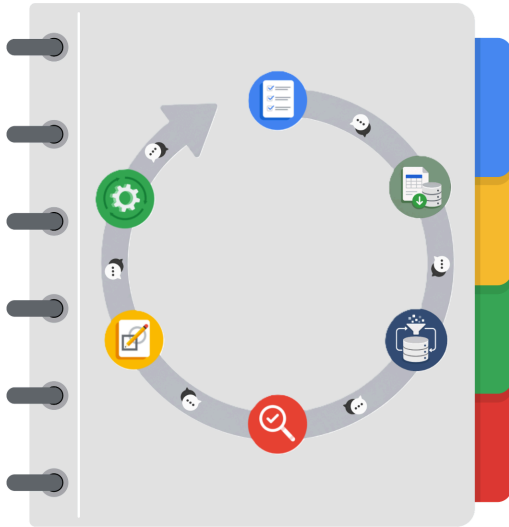
Portfolio Project Recap

Many of the goals I accomplished throughout the individual course portfolio projects are integrated into this end-to-end data analysis capstone project. These include:

- Creating a clear and structured project proposal.
- Demonstrating my understanding of SQL and relational database concepts.
- Using SQL to ingest, explore, clean, transform, validate, and organize data through structured queries and database operations.
- Organizing and analyzing a large-scale dataset to uncover meaningful insights and communicate the underlying business story.
- Developing a series of SQL scripts for ELT, feature engineering, data validation, and analytical querying, alongside a DuckDB-powered Jupyter Notebook documenting the complete APPASA methodology and reproducing the analytical workflow through SQL queries and visualizations.
- Computing descriptive statistics, performing exploratory and business-focused analyses, and creating analytical visualizations to support data interpretation and executive decision-making.
- Communicating findings through an executive summary, an interactive Power BI report, strategic business recommendations, and implementation-focused action plans for stakeholders.
- Translating analytical findings into evidence-based business recommendations, implementation strategies, success metrics, and future analytical opportunities through the final Act stage of the APPASA framework.

This capstone brings together the knowledge and technical skills I developed throughout the program, demonstrating my ability to execute a complete end-to-end data analysis project, from business problem definition to actionable business recommendations.

THE APPASA WORKFLOW



I will demonstrate to the company's marketing analytics team how I would apply the APPASA workflow to the upcoming Cyclistic Bike Share project. For each question presented in this APPASA strategy document, I will provide a structured and thoughtful response aligned with the corresponding stage of the APPASA framework: Ask, Prepare, Process, Analyze, Share and Act. This approach ensures clarity in my methodology, highlights my strategic planning skills, and illustrates a well-organized path from problem understanding to actionable insights.

Project Tasks

The following questions have been identified as essential to the Member vs Casual Rider Behavior Analysis Project. These questions are primarily situated within the Act stage of the APPASA workflow, focusing on interpreting analytical findings, developing evidence-based business recommendations, evaluating their potential impact, and identifying future opportunities. I will address each question by drawing on relevant materials from the SQL scripts, project notebook, the APPASA framework, dashboard, and best practices in communicating data-driven business recommendations.





Data Project Questions & Considerations

Business Conclusions

- What is the overall conclusion of this analysis?

The analysis concludes that annual members and casual riders exhibit fundamentally different usage patterns, reflecting distinct transportation needs and motivations. Annual members primarily use Cyclistic bicycles for routine commuting, characterized by shorter ride durations, consistent weekday usage, and pronounced peak-hour travel. In contrast, casual riders demonstrate longer, leisure-oriented trips, stronger weekend activity, and increased ridership during the warmer months.

These behavioural differences indicate that casual riders represent the greatest opportunity for membership growth. Rather than attempting to alter rider behaviour, Cyclistic should align its marketing strategy with existing recreational usage patterns by promoting membership options and incentives that appeal to high-engagement casual riders. By targeting this segment with tailored value propositions and flexible membership offerings, Cyclistic can increase annual memberships while leveraging established rider behaviour patterns.

- Which findings have the greatest business impact?

Three findings have the greatest business impact because they directly inform strategies for increasing annual memberships. First, casual riders consistently take longer, leisure-oriented trips and are disproportionately represented in extended ride duration categories, indicating that they primarily use Cyclistic for recreation rather than routine transportation. Second, annual members exhibit predictable weekday commuting patterns with shorter ride durations and pronounced peak-hour usage, demonstrating that memberships currently deliver the greatest value to regular commuters. Third, casual rider activity increases substantially during weekends and the warmer months, creating clear seasonal opportunities to engage highly active riders when they are most likely to consider membership.

Together, these findings suggest that the most effective growth strategy is not to change how casual riders use the service, but to design membership offerings and marketing campaigns that align with their existing recreational riding behaviour. This insight provides a clear, evidence-based direction for Cyclistic to increase annual memberships by targeting high-engagement casual riders with tailored promotional strategies and flexible membership options.



Business Recommendations

- Why was promoting leisure-oriented membership value recommended?

Promoting leisure-oriented membership value was recommended because the analysis demonstrated that casual riders primarily use Cyclistic bicycles for recreational purposes rather than routine commuting. Casual riders consistently recorded longer average ride durations, were more heavily represented in extended ride duration categories, and exhibited stronger weekend and seasonal riding patterns. These behavioural characteristics indicate that their motivations differ substantially from those of annual members.

As a result, marketing memberships primarily as a commuting solution is unlikely to resonate with this audience. Instead, emphasizing benefits that support recreational riding, such as extended ride value, weekend-focused rewards, scenic route campaigns, and leisure-oriented promotions, better aligns the membership offering with existing rider behaviour. By positioning annual memberships as a valuable option for frequent recreational riders, Cyclistic can increase the attractiveness of membership without requiring casual riders to change how they use the service.

- Why was reducing commitment barriers recommended?

Reducing commitment barriers was recommended because the analysis indicates that many casual riders use Cyclistic frequently enough to represent strong membership prospects but may be discouraged by the perceived commitment of purchasing an annual membership. Their recurring recreational usage suggests continued engagement with the service, yet the preference for single-ride or day-pass options implies that the existing membership structure may not adequately address their needs or purchasing preferences.

Introducing lower-commitment membership options, trial memberships, limited-time introductory offers, or promotional incentives could encourage casual riders to experience the benefits of membership with reduced perceived risk. By lowering the barriers to entry, Cyclistic can increase the likelihood that engaged casual riders transition to annual memberships while supporting the company's objective of expanding its long-term member base.

- Why were seasonal and peak conversion campaigns recommended?

Seasonal and peak conversion campaigns were recommended because the analysis revealed that casual rider activity increases significantly during weekends and the warmer months, when recreational cycling demand is highest. These periods represent times of increased engagement with the Cyclistic service, making casual riders more receptive to membership-related marketing and promotional offers.

By concentrating marketing efforts during these high-activity periods, Cyclistic can reach riders when they are experiencing the greatest value from the service and are more likely to recognize the benefits of an annual membership. Aligning campaigns with periods of peak usage enables the company to maximize marketing



effectiveness, improve conversion opportunities, and support its objective of increasing annual memberships through timely, data-driven customer engagement.

- How do the recommendations support Cyclistic's objective of increasing annual memberships?

The recommendations directly support Cyclistic's objective of increasing annual memberships by aligning marketing strategies with the observed behaviour and preferences of casual riders. Rather than applying broad, generic promotions, each recommendation addresses a specific behavioural pattern identified during the analysis. Promoting leisure-oriented membership value aligns with the recreational usage patterns of casual riders, reducing commitment barriers makes membership more accessible to engaged riders who may be hesitant to commit to an annual plan, and seasonal conversion campaigns target periods when casual rider activity and engagement are at their highest.

Together, these recommendations provide a targeted, data-driven approach to converting existing casual riders into annual members. By focusing on the rider segment with the greatest conversion potential and tailoring marketing initiatives to their demonstrated behaviour, Cyclistic can improve the effectiveness of its membership campaigns and support sustainable long-term growth.

- How are the recommendations supported by the analytical findings?

The recommendations are directly supported by the analytical findings identified throughout the analysis. The recommendation to promote leisure-oriented membership value is supported by evidence that casual riders primarily engage in longer, recreational trips and demonstrate stronger weekend usage patterns. Reducing commitment barriers is supported by the finding that casual riders use the service regularly but continue to rely on single-ride or short-term purchase options, suggesting an opportunity to encourage membership through more accessible entry points. Seasonal and peak conversion campaigns are supported by the observed increase in casual rider activity during the warmer months and periods of higher recreational demand.

Because each recommendation is derived from measurable behavioural trends rather than assumptions, they provide an evidence-based strategy for increasing annual memberships. This alignment between analytical findings and business recommendations ensures that Cyclistic's marketing initiatives address the characteristics and motivations of the rider segment with the greatest conversion potential.



Business Impact & Next Steps

- How can Cyclistic apply these insights?

Cyclistic can apply these insights by developing marketing strategies and membership offerings that reflect the behavioural patterns identified throughout the analysis. Rather than using a one-size-fits-all approach, the company should focus its efforts on high-engagement casual riders by promoting membership benefits that align with recreational cycling, introducing lower-commitment membership options, and launching targeted campaigns during weekends and peak riding seasons when casual rider activity is highest.

In addition, Cyclistic can use these insights to guide future marketing decisions, allocate promotional resources more effectively, and continuously evaluate campaign performance using key metrics such as membership conversion rates, customer acquisition costs, and member retention. Applying data-driven strategies based on observed rider behaviour enables the company to improve the effectiveness of its marketing initiatives while supporting the long-term objective of increasing annual memberships.

- What actions should the marketing team take next?

The marketing team should develop and implement targeted campaigns that focus on converting high-engagement casual riders into annual members. This includes promoting leisure-oriented membership benefits, introducing limited-time promotional offers or trial memberships, and scheduling campaigns during weekends and peak riding seasons when casual rider activity is highest. Marketing messages should emphasize the value of membership for frequent recreational riders rather than focusing exclusively on commuting benefits.

Following implementation, the marketing team should continuously monitor campaign performance using metrics such as membership conversion rates, campaign engagement, customer acquisition costs, and member retention. Regular evaluation of these metrics will enable the team to refine marketing strategies, optimize promotional efforts, and ensure that future campaigns remain aligned with rider behaviour and Cyclistic's objective of increasing annual memberships.

- What actions should the executive team take?

The executive team should use the findings from this analysis to guide strategic decisions related to membership growth, marketing investment, and product development. This includes prioritising resources toward data-driven marketing initiatives, supporting the introduction of flexible membership options that address the needs of casual riders, and encouraging cross-functional collaboration between marketing, operations, and customer experience teams to implement the recommended strategies effectively.

In addition, the executive team should establish clear performance objectives and regularly review key business metrics, including annual membership growth, conversion rates, customer acquisition costs, and member retention. Monitoring these outcomes will enable leadership to evaluate the effectiveness of the



implemented strategies, make informed adjustments when necessary, and ensure that organisational efforts remain aligned with Cyclistic's long-term objective of increasing annual memberships.

- How could the success of these recommendations be measured?

The success of these recommendations can be measured by monitoring key performance indicators that reflect the effectiveness of Cyclistic's membership growth strategy. Primary metrics should include the conversion rate of casual riders to annual members, overall annual membership growth, customer acquisition costs, member retention rates, and the return on investment (ROI) of marketing campaigns. Tracking these indicators over time enables Cyclistic to determine whether the recommended strategies are successfully increasing membership adoption while maintaining cost-effective marketing performance.

In addition, Cyclistic should compare these metrics before and after implementing the recommended initiatives and continuously evaluate campaign performance during peak riding seasons. Regular performance reviews will help identify which strategies generate the highest conversion rates, allowing the company to refine future marketing efforts and support sustainable long-term membership growth.

Future Opportunities

- What additional data could strengthen future analyses?

Future analyses could be strengthened by incorporating additional customer and operational data that provides deeper insight into rider behaviour and membership conversion. Information such as rider demographics, membership purchase history, trip purpose, marketing campaign interactions, customer feedback, and promotional offer redemption would enable more detailed segmentation and help identify the factors that most strongly influence membership decisions. Geographic information, station capacity data, weather conditions, and special event schedules could also improve the understanding of external factors affecting ridership patterns.

Integrating these additional data sources would support more comprehensive behavioural analysis, enable more precise targeting of potential members, and improve the evaluation of marketing effectiveness. Access to richer customer and operational data would allow Cyclistic to develop more personalised, evidence-based strategies for increasing annual memberships and enhancing the overall customer experience.

- What future analyses could build upon this project?

Future analyses could build upon this project by investigating the long-term effectiveness of membership conversion strategies and evaluating how rider behaviour changes after casual riders become annual members. Comparative studies across multiple years could identify evolving seasonal trends, changes in rider preferences, and the sustained impact of marketing initiatives on membership growth. Additional analyses



could also examine station-level demand, geographic riding patterns, and customer segmentation to support more targeted operational and marketing decisions.

Furthermore, predictive analytics and machine learning models could be developed to identify casual riders with the highest likelihood of purchasing an annual membership based on their historical riding behaviour. These advanced analytical approaches would enable Cyclistic to proactively target high-potential customers, optimise marketing resource allocation, and further enhance data-driven decision-making for future membership growth initiatives.

- What limitations should stakeholders consider when implementing these recommendations?

Stakeholders should recognise that these recommendations are based on historical trip data and behavioural patterns observed during the analysis period. Rider preferences, travel demand, market conditions, and external factors such as weather, local events, infrastructure changes, or economic conditions may change over time, potentially influencing the effectiveness of the recommended strategies. Additionally, the analysis was based primarily on trip-level operational data and did not include customer demographics, marketing interactions, or qualitative feedback that could provide deeper insight into the motivations behind membership decisions.

Consequently, the recommendations should be viewed as evidence-based strategic guidance rather than guaranteed outcomes. Cyclistic should implement these initiatives alongside continuous performance monitoring and regularly evaluate key business metrics to assess their effectiveness. Ongoing data collection and periodic reanalysis will enable the company to refine its strategies, adapt to changing rider behaviour, and support sustained growth in annual memberships.